Budget and Justification

Project Title: Phytochemical characterization of macroalgae for food and high value products (PhycoPRO)

LINE ITEM BUDGET FOR UP-MSI
TOTAL PROPOSED BUDGET FOR 2 YEARS: 110,223,004.80

A. Personnel: List employee(s) (including names for each identified position) of the applicant/recipient organization, including in-kind costs for those positions whose work is tied to the grant project.+

Personnel for PH Component: UP-MSI

Position	Name	Annual Salary/Rate	Level of Effort	Cost for YR 1	Cost for YR 2
(1) Project Leader	Michael Y. Roleda	105,600.00	30%	105,600.00	105,600.00
(2) Co-Project Leader	Irene B. Rodriguez	90,000.00	30%	90,000.00	90,000.00
(3) Project Staff	Lilibeth Salvador-Reyes	90,000.00	10%	90,000.00	90,000.00
(4-9) SRS II	To be selected	3,926,160.00	100%	3,926,160.00	3,926,160.00
(10-11) URA II	To be selected	930,840.00	100%	930,840.00	930,840.00
(12-13) Project Assistant 1	To be selected	547,152.00	100%	547,152.00	547,152.00
		TOTAL	**PhP**	**5,059,752.00**	**5,059,752.00**

JUSTIFICATION: Describe the role and responsibilities of each position.

(1) and (2): The Project Leader and Co-Project Leader will provide general research directions, oversight of the project, supervision of research staff, and will be considered key staff. The Project Leaders will be paid honoraria.

(3) : The Project Staff will provide guidance and suggestions for successful implementation of the grant and will be considered key staff. The Project Staff will be paid honoraria.

(4-9) : Six (6) Science Research Specialist II (SRS II) are MS graduates and will carry out majority of the experiments and will be considered key staff. Each SRS II will be in-charge of specific task or research target including method optimization, seaweed culturing, bio-assays, chemical extraction, and other research tasks.

(10-11): Two University Research Associate II (URA II) are MS students and will carry out majority of the experiments and will be considered key staff. Each URA II will be in-charge of specific task or research target including method optimization, seaweed culturing, bio-assays, chemical extraction, and other research tasks.

(12-13): Two Project Assistants will assist in document processing including licenses, bids and awards, assist in sampling activities, hatchery cultures, preparation of seaweed voucher specimens, and laboratory activities.

C. Travel (for Philippines side only): UP-MSI

Travel expenses for PH counterpart

Item(s)	Cost for YR 1	Cost for YR 2
Travel Expenses		
Sampling	1,755,000.00	1,298,000.00
Travel expenses for establishing network with local government units	785,000.00	640,000.00
Travel expenses for workshops	785,000.00	640,000.00
Travel insurance for staff	75,000.00	75,000.00
TOTAL	**3,400,000.00**	**2,653,000.00**

Sampling costs: The sampling cost will cover airfare and per diem costs. The sampling will involve number of participants and number of days that will be unique to the requirements of the study site.

	YEAR 1	YEAR 2
Caramoan, Camarines Sur:	199,000.00	
Dinagat, Surigao Del Norte:		189,000.00
Tandubas, Tawi Tawi:		254,000.00
Batan, Batanes:	464,000.00	
Bolinao, Pangasinan:	100,000.00	100,000.00
Currimao, Ilocos Norte:	100,000.00	100,000.00
Guiuan, Eastern Samar:	169,000.00	
Biri, Northern Samar:		219,000.00
Iba, Zambales:		112,000.00
Busuanga, Palawan:	212,000.00	
Sorsogon, Sorsogon:		162,000.00
Legazpi City, Albay:	169,000.00	
Bien Unido, Bohol:	180,000.00	
Lian, Batangas:	162,000.00	162,000.00
TOTAL	1,755,000.00	1,298,000.00

Travel Expenses for Sampling:

To Caramoan, Camarines Sur: 199,000.00 [airfare and per diem during travel = 90,000; per diem during sampling = 1500/day; 1,500 x 10 people x 7 days = 105,000.00; incidentals = 4,000.00]

To Dinagat, Surigao Del Norte: 189,000.00 [airfare and per diem during travel = 80,000; per diem during sampling = 1500/day; 1,500 x 10 people x 7 days = 105,000.00; incidentals = 4,000.00]

To Tandubas, Tawi Tawi: 254,000.00 [airfare = 100,000; per diem = 1500/day; 1,500 x 10 x people x 10 days = 150,000.00; incidentals = 4,000.00]

To Batan, Batanes: 464,000.00 [airfare = 310,000; per diem = 1500/day; 1,500 x 10 x people x 10 days = 150,000.00; incidentals = 4,000.00]

To Bolinao, Pangasinan: 100,000.00 [per diem = 1500/day; 1500 x 8 people x 8 days = 96,000.00; incidentals = 4,000.00]

To Currimao, Ilocos Norte: 100,000.00 [per diem = 1500/day; 1500 x 8 people x 8 days = 96,000.00; incidentals = 4,000.00]

To Guiuan, Eastern Samar: 169,000.00 [airfare and per diem during travel = 60,000; per diem during sampling = 1500/day; 1,500 x 10 people x 7 days = 105,000.00; incidentals = 4,000.00]

To Biri, Northern Samar: 219,000.00 [airfare and per diem during travel = 110,000; per diem during sampling = 1500/day; 1,500 x 10 people x 7 days = 105,000.00; incidentals = 4,000.00]

To Iba, Zambales: 112,000.00 [per diem = 1500/day; 1500 x 8 people x 9 days = 108,000.00; incidentals = 4,000.00]

To Busuanga, Palawan: 212,000.00 [airfare and per diem during travel = 54,000; per diem during sampling = 2200/day; 2200 x 10 people x 7 days = 154,000.00; incidentals = 4,000.00]

To Sorsogon, Sorsogon: 162,000.00 [airfare and per diem during travel = 50,000; per diem = 1500/day; 1500 x 8 people x 9 days = 108,000.00; incidentals = 4,000.00]

To Legazpi, Albay: 169,000.00 [airfare and per diem during travel = 60,000; per diem during sampling = 1500/day; 1,500 x 10 people x 7 days = 105,000.00; incidentals = 4,000.00]

To Bien Unido, Bohol: 180,000.00 [airfare and per diem during travel = 50,000; per diem during sampling = 1800/day; 1,800 x 10 people x 7 days = 126,000.00; incidentals = 4,000.00]

To Lian, Batangas: 162,000.00 [per diem = 2200/day; 2200 x 8 people x 9 days = 158,400.00; incidentals = 3,600.00]

NOTE: In most of the sampling trips, the researchers will be availing of the MSI vehicle and will entail services of an MSI driver, which will make the number of people equal to 10.

Travel Expenses for Establishing LGU network/Workshops:

Year 1

To Caramoan, Camarines Sur: 169,000.00 [airfare and per diem during travel = 90,000; per diem during workshop = 1500/day; 1,500 x 10 people x 5 days = 75,000.00; incidentals = 4,000.00]

To Dinagat, Surigao Del Norte: 159,000.00 [airfare and per diem during travel = 80,000; per diem during workshop = 1500/day; 1,500 x 10 people x 5 days = 75,000.00; incidentals = 4,000.00]

To Tandubas, Tawi Tawi: 179,000.00 [airfare = 100,000; per diem = 1500/day; 1,500 x 10 x people x 5 days = 75,000.00; incidentals = 4,000.00]

To Guiuan, Eastern Samar: 139,000.00 [airfare and per diem during travel = 60,000; per diem during workshop = 1500/day; 1,500 x 10 people x 5 days = 75,000.00; incidentals = 4,000.00]

To Legazpi, Albay: 139,000.00 [airfare and per diem during travel = 60,000; per diem during workshop = 1500/day; 1,500 x 10 people x 5 days = 75,000.00; incidentals = 4,000.00]

Year 2

To Caramoan, Camarines Sur: 169,000.00 [airfare and per diem during travel = 90,000; per diem during workshop = 1500/day; 1,500 x 10 people x 5 days = 75,000.00; incidentals = 4,000.00]

To Biri, Northern Samar: 189,000.00 [airfare and per diem during travel = 110,000; per diem during workshop = 1500/day; 1,500 x 10 people x 5 days = 75,000.00; incidentals = 4,000.00]

To Busuanga, Palawan: 168,000.00 [airfare and per diem during travel = 54,000; per diem during workshop = 2200/day; 2200 x 10 people x 5 days = 110,000.00; incidentals = 4,000.00]

To Sorsogon, Sorsogon: 114,000.00 [airfare and per diem during travel = 50,000; per diem = 1500/day; 1500 x 8 people x 5 days = 60,000.00; incidentals = 4,000.00]

Travel insurance for staff

Insurance for research staff participating in sampling trips will be applied for. With the current situation of the COVID-19 pandemic, insurance covering COVID-19 and dengue will also be availed of aside from the travel insurance. This is estimated to be around PhP 7,500 per year for each research staff.

D.Equipment (for Philippines-side only): UP-MSI

Item(s)	Cost/unit (PhP)	YR 1		YR 2	
		No. of Units	Sub-Total for YR 1	No. of Units	Sub-Total for YR 2
graphite furnace atomic absoprtion spectrophotometer with complete parts and accessories	4,500,000.00	1	4,500,000.00		
CHNS/O elemental analyzer with complete parts and accessories	4,500,000.00	1	4,500,000.00		
microwave digestion system with complete parts and accessories	4,000,000.00	1	4,000,000.00		
fumehoods with complete parts and accessories	1,000,000.00	1	1,000,000.00		
GCMS and accessories	11,000,000.00	1	11,000,000.00		

HPLC with DAD detector and accessories	3,500,000.00	1	3,500,000.00		
HPLC MS and accessories	20,000,000.00	1	20,000,000.00		
lyophilizer and accessories	1,200,000.00	1	1,200,000.00		
ultralow freezer and accessories	1,300,000.00	2	2,600,000.00		
UV-Vis spectrophotometer and accessories	700,000.00	1	700,000.00		
Multi-channel peristaltic pump and accessories	300,000.00	2	600,000.00		
rotary evaporator and accessories	500,000.00	2	1,000,000.00		
Oven drier and accessories	200,000.00	1	200,000.00		
analytical balance and accessories	100,000.00	2	200,000.00		
toploading balance and accessories	50,000.00	2	100,000.00		
High resolution camera and accessories for seaweed documentation	100,000.00	1	100,000.00		
Temperature and humidity data logger and accessories	70,000.00	4	280,000.00		
hot plate/magnetic stirrer and accessories	50,000.00	4	200,000.00		
Desktops/laptops and accessories	50,000.00	6	300,000.00		
Furnace and accessories	450,000.00	1	450,000.00		
Sonicator and accessories	500,000.00	2	1,000,000.00		
vacuum pump and accessories	200,000.00	2	400,000.00		
pH meter, microwave oven, micro-centrifuge, vortex mixer, and accessories	170,000.00	1	170,000.00		
qPCR, gradient PCR, and accessories	3,500,000.00	1	3,500,000.00		
refrigerated centrifuge with different rotors and accessories	1,500,000.00	1	1,500,000.00		
water bath with temperature control and accessories	475,000.00	2	950,000.00		
multiplate reader and accessories	2,000,000.00	1	2,000,000.00		
multiparameter and accessories for water analysis	400,000.00	2	800,000.00		
TOTAL (PhP)			**66,750,000.00**		**0.00**

JUSTIFICATION: Describe the need or purpose of the equipment (e.g. how each unit will be used). UP-MSI

Equipment	Purpose	User	Location of the equipment
graphite furnace atomic absorption spectrophotometer with complete parts and accessories	For multi-elemental analysis of trace metals in seaweed samples and seaweed extracts.	For use of the research staff during the execution of the grant and will become part of the *Seaweed Chemistry facility*. Will be accessed by collaborating HEIS through a MOA that will be executed.	Will be housed and maintained at UP-MSI.
CHNS/O elemental analyzer with complete parts and accessories	For the determination of particulate organic carbon and nitrogen content of seaweed samples and extracts.	For use of the research staff during the execution of the grant and will become part of the *Seaweed Chemistry facility*.	Will be housed and maintained at UP-MSI.

microwave digestion system with complete parts and accessories	Will be used for the sample processing of seaweed samples and extracts to make them suitable for trace metal determination.	For use of the research staff during the execution of the grant and will become part of the *Seaweed Chemistry facility*.	Will be housed and maintained at UP-MSI.
fumehoods with complete parts and accessories	Will be used for the safe processing of the samples to ensure protection of research staff.	For use of the research staff during the execution of the grant and will become part of the *Seaweed Chemistry facility*.	Will be housed and maintained at UP-MSI.
GCMS and accessories	For analysis of fatty acid methyl esters (FAMEs), derivatized iodine-containing compounds, and thermally-labile compounds of interest.	For use of the research staff during the execution of the grant and will become part of the *Seaweed Chemistry facility*. Will be accessed by collaborating HEIS through a MOA that will be executed.	Will be housed and maintained at UP-MSI.
HPLC-DAD detector and accessories	For analysis of amino acids, fucoxanthin and fucosterol, β-carotene, and other thermally-stable compounds of interest.	For use of the research staff during the execution of the grant and will become part of the *Seaweed Chemistry facility*.	Will be housed and maintained at UP-MSI.
HPLC-MS and accessories	For analysis of small molecules of interest and other thermally-unstable compounds of interest.	For use of the research staff during the execution of the grant and will become part of the *Seaweed Chemistry facility*. Will be accessed by collaborating HEIS through a MOA that will be executed.	Will be housed and maintained at UP-MSI.
Lyophilizer and accessories	Required for sample preparation and pre-treatment.	For use of the research staff during the execution of the grant and will become part of the *Seaweed Chemistry facility*	Will be housed and maintained at UP-MSI.
Ultralow freezer and accessories	For use in preservation and storage of samples and extracts.	For use of the research staff during the execution of the grant and will become part of the *Seaweed Chemistry facility*	Will be housed and maintained at UP-MSI.
UV-Vis spectrophotometer and accessories	For analysis of phenolic content, determination of absorption spectra, and relevant antioxidant assays.	For use of the research staff during the execution of the grant and will become part of the *Seaweed Chemistry facility*	Will be housed and maintained at UP-MSI.
Multi-channel peristaltic pump and accessories	Required for volume-controlled and automated	For use of the research staff during the execution of the grant and will be utilized by	Will be housed and

	mixing of reagents in antioxidant assays	the *Seaweed Chemistry facility*.	maintained at UP-MSI.
Rotary evaporator and accessories	Required for extraction of target compounds from samples.	For use of the research staff during the execution of the grant and will be utilized by the *Seaweed Chemistry facility*.	Will be housed and maintained at UP-MSI.
Oven drier and accessories	Required for sample preparation and pre-treatment.	For use of the research staff during the execution of the grant and will become part of the *Seaweed Chemistry facility*.	Will be housed and maintained at UP-MSI.
Analytical balance and accessories	Required for sample preparation and pre-treatment.	For use of the research staff during the execution of the grant and will be utilized by the *Seaweed Chemistry facility*.	Will be housed and maintained at UP-MSI.
Toploading balance and accessories	Required for sample preparation and pre-treatment.	For use of the research staff during the execution of the grant and will be utilized by the *Seaweed Chemistry facility*.	Will be housed and maintained at UP-MSI.
High resolution camera and accessories	Required for sample documentation.	For use of the research staff during the execution of the grant and will be utilized by the *Seaweed Chemistry facility*.	Will be housed and maintained at UP-MSI.
Temperature and humidity data logger and accessories	Required for recording of temperature and humidity.	For use of the research staff during the execution of the grant and will be utilized by the *Seaweed Chemistry facility*.	Will be housed and maintained at UP-MSI.
Hot plate and magnetic stirrer and accessories	Required for sample preparation and pre-treatment.	For use of the research staff during the execution of the grant and will be utilized by the *Seaweed Chemistry facility*.	Will be housed and maintained at UP-MSI.
Desktops/laptops and accessories	Required for documentation, data analysis, data processing, and writing.	For use of the research staff during the execution of the grant and will be utilized by the *Seaweed Chemistry facility*.	Will be housed and maintained at UP-MSI.
Furnace and accessories	Required for ashing of seaweed raw materials and products.	For use of the research staff during the execution of the grant and will be utilized by the *Seaweed Chemistry facility*.	Will be housed and maintained at UP-MSI.
Sonicator and accessories	Required for cleaning and extraction purposes.	For use of the research staff during the execution of the grant and will be utilized by the *Seaweed Chemistry facility*.	Will be housed and maintained at UP-MSI.
Vacuum pump and accessories	Required for filtration and extraction processes.	For use of the research staff during the execution of the grant and will be utilized by the *Seaweed Chemistry facility*.	Will be housed and maintained at UP-MSI.
pH meter, microwave oven, microcentrifuge,	Required for routine pH determination and other laboratory procedures.	For use of the research staff during the execution of the grant and will be utilized by	Will be housed and maintained at UP-MSI.

vortex mixer, and accessories		the *Seaweed Chemistry facility*.	
qPCR, gradient PCR, and accessories	Will be used for targeted transcriptomics analyses.	For use of the research staff during the execution of the grant and will be utilized by the *Seaweed Chemistry facility*. Will be accessed by collaborating HEIS through a MOA that will be executed.	Will be housed and maintained at UP-MSI.
refrigerated centrifuge with different rotors and accessories	Will be used for samples processing for pre-concentration or separation purposes.	For use of the research staff during the execution of the grant and will be utilized by the *Seaweed Chemistry facility*.	Will be housed and maintained at UP-MSI.
water bath with temperature control and accessories	Will be used for experiments requiring precise control of temperature during processing.	For use of the research staff during the execution of the grant and will be utilized by the *Seaweed Chemistry facility*.	Will be housed and maintained at UP-MSI.
multiplate reader and accessories	Will be used for biochemical assays with samples that are available at minimum volumes and require almost simultaneous measurements.	For use of the research staff during the execution of the grant and will be utilized by the *Seaweed Chemistry facility*.	Will be housed and maintained at UP-MSI.
multiparameter and accessories for water analysis	Will be used for routine measurements of physico-chemical parameters during sampling.	For use of the research staff during the execution of the grant and will be utilized by the *Seaweed Chemistry facility*.	Will be housed and maintained at UP-MSI.

After the end of the grant, most of the equipment acquired through this project will constitute the *Seaweed Chemistry facility* housing sophisticated equipment. A close coordination with the UP-MSI will be sought to address maintenance and operation costs for the equipment. For common use equipment, all MSI research staff may use the equipment provided that operators will undergo sufficient training and agree to shoulder technical fees that may be incurred if their usage of specific equipment led to equipment malfunction. For sophisticated equipment, the use may be made accessible to other researchers after training and fees will be collected to cover cost of chemicals and other requirements. The core equipment facility will be supervised by the Project Leaders of this project with the assistance of other UP-MSI faculty with relevant background.

Is the host institution willing to maintain the equipment/facility?

_✓_ YES.

If YES, provide means by which the equipment/facility will be maintained.

The equipment will be maintained in the Seaweed Chemistry Facility at the UP-MSI.

__ NO

E. Supplies and Materials:
Supplies and Materials for PH side: UP-MSI

Item(s)	Cost/unit (PhP)	YR 1		YR 2	
		No. of units	Sub-Total for YR 1	No. of units	Sub-Total for YR 1
Extraction solvents	~6,000.00	~20	100,000.00	~40	200,000.00
HPLC grade solvents	~16,500.00	~9	150,000.00	~9	150,000.00
Certified reference materials	~30,000.00	~9	250,000.00	~4	100,000.00
ICPMS single element standards	~8,000.00	~32	250,000.00	~6	50,000.00
High purity acids and high purity gases (He, Ar, N_2, air)	~10,000.00	~15	150,000.00	~20	200,000.00
General laboratory chemicals	~9,000.00	~18	150,000.00	~18	150,000.00
Chemicals for preparation of artificial seawater and other required reagents	~1,000.00 to 5,000.00	~15	100,000.00	~25	150,000.00
Laboratory glassware/plastic ware and other materials	~500.00 to 10,000.00	1 to 50 (various)	250,000.00	1 to 20 (various)	100,000.00
TOTAL (PhP)			**1,400,000.00**		**1,100,000.00**

JUSTIFICATION:

Extraction solvents: the solvents that will be used in this work include hexane, toluene, methanol, ethanol, diethyl ether, and other solvents that may be needed for harsher extraction conditions. The average cost of hexane, toluene, and diethyl ether is about PhP 6,000.00 per 2L while the average cost of methanol, ethanol, and other alcohol-containing solvent is about PhP 6,500.00 per 4L. It is expected that significant volumes of extraction solvents will be utilized for the duration of this project.

HPLC-grade solvents: elution solvents for HPLC analysis will include methanol, acetonitrile, dichloromethane, and other solvents that may be necessary. The average cost of HPLC-grade solvents is PhP 16,500.00 per 8L and significant volumes will be utilized for optimization and validation of methods and analysis of seaweed samples.

Certified reference materials: optimization and validation of standard methods will require the use of certified reference materials with matrices that are closest to the real samples. The average cost of reference materials is about PhP 30,000.00 including freight costs.

ICPMS single element standards: single element standards will be used for preparation of standard calibration standard solution of target trace metal analytes and will be continually used for evaluation of the standard procedures and in the actual sample analyses.

High purity acids: high purity nitric acid, hydrochloric acid, hydrofluoric acid, acetic acid, and sulfuric acid will be required in this project for the complete mineralization of seaweed and seaweed extracts through acid-assisted hot plate digestion. In addition, high purity acids will also be used in the preparation of ICPMS standards and samples for analysis.

High purity gases: high purity gases such as argon, nitrogen, helium, and oxygen will be required for the operation of the ICPMS (argon, helium, nitrogen, and oxygen) and GC-MS (helium).

Chemicals for preparation of artificial seawater and other reagents will be required in keeping seaweed species alive in culture when samples are required for subsequent validation of physicochemical parameters.

Laboratory glassware/plastic ware and other materials: this project will require use of both glass and plastic ware for solution and sample preparation. For all work focusing on trace metal analysis and speciation, plastic ware will be used while glass will be used for all other experiments.

F. Professional Services: Services of a consultant in bookkeeping will be sought whenever necessary and services of a lawyer will also be sought when the need arises (e.g., when purchase of equipment is problematic due to unwanted practices of suppliers).

COSTS FOR CONTRACTS MUST BE BROKEN DOWN IN DETAIL AND NARRATIVE JUSTIFICATION. IF APPLICABLE, NUMBERS OF CLIENTS SHOULD BE INCLUDED IN THE COSTS.

Name	Service	Rate	Cost for YR —	Cost for YR —
To be identified	Bookkeeping	SG 11 (30,527/mo)	366,324.00	366,324.00
To be identified	Daily labor support	400/day	33,676.00	33,676.00
		TOTAL	400,000.00	400,000.00

JUSTIFICATION: Explain the need for each contractual agreement and how they relate to the overall project.

1. The services of a bookkeeper is needed because this is a big grant that will require timely submission of financial and technical reports. The processing of vouchers for procurement, payment or travel of research staff will require the services of the bookkeeper.
2. The remaining funds will be allotted for minimum wages (daily labor costs) for daily wage earners that will be hired for services rendered like assistance in handling heavy sampling containers, cleaning of hatchery spaces/tanks, and other manual tasks that the research staff will need assistance for.

G. Construction: NOT ALLOWED

H. Other: expenses not covered in any of the previous budget categories

OTHER EXPENSES FOR THE PH COUNTERPART: UP-MSI

Item(s)	Cost for YR 1	Cost for YR 2
Communication expenses		
Communication expenses during sampling	20,000.00	20,000.00
Repairs and Maintenance of Facilities		
Maintenance of clean room	20,000.00	30,000.00
Renovation cost	500,000.00	
Transportation and Delivery Expenses		
Freight costs	150,000.00	150,000.00
Training Expenses		
Workshop in collaboration with Sorsogon State University		95,000.00

Workshop in collaboration with MSU Naawan		90,000.00
Workshop in collaboration with University of Eastern Philippines		90,000.00
Workshop in collaboration with Partido State University		90,000.00
Seaweed Farmers Training/Workshops	14,000,000.00	
Membership Dues for the Institution		
		-
Rent Expenses		
Boats for sampling	50,000.00	200,000.00
Representation Expenses		
Annual meetings (food expenses)	15,000.00	15,000.00
Kickoff meeting (food expenses)	15,000.00	
Project evaluation meeting (food expenses)		15,000.00
Subscription expenses		
Purchase of access to relevant scientific articles	20,000.00	50,000.00
Professional fees		
Consultant	400,000.00	400,000.00
Taxes, Insurance Premiums, and Other Fees		
Environmental fees for sampling	50,000.00	50,000.00
Chemical waste treatment	50,000.00	100,000.00
Other Maintenance and Operating Expenses		
Instrument fee for inter-calibration purposes	60,000.00	60,000.00
Publication expenses	250,000.00	400,000.00
TOTAL	**15,600,000.00**	**1,855,000.00**

JUSTIFICATION:

Communication expenses will cover cost for purchasing of SIM and prepaid cards for use of project staff in coordinating sampling activities.

Maintenance of clean room will include costs of changing eap filters every quarter during the operation of a clean room. Renovation cost will cover costs of room preparation for the sophisticated equipment that will be purchased through this project. The three MS equipment (ICPMS, HPLC-MS, GC-MS) will require specific conditions for the optimum performance of the equipment. HPLC-MS and GC-MS will need dedicated rooms and the ICPMS will require a class 1000 or better clean conditions. In addition to this, electrical supply, water and gas lines, and provisions for fume hoods in rooms housing specific equipment will also need to be properly installed.

Freight costs will cover fees for sending samples to the CA counterpart and costs for sending samples from within the country whenever the need arises, e.g., from Tawi Tawi to QC.

Water and electricity costs will cover fees that may be incurred during sampling especially in remote areas where use of generators may be required.

Training Expenses – the funds will cover the cost of workshop materials and food during the duration of the workshops in the partner HEIs and in UP-MSI. Th

Seaweed Farmers Training/Workshops– the funds will cover (1) Farmer training (20-30 farmers per village in 5 villages per hub) focusing on practices that can minimise losses from disease and epiphytes and reduces marine plastic pollution. Costs will cover community organising/preparation for the workshops, seaweed farmer mobilisation, resource personnel to deliver training, training venue, materials, snacks/refreshments for participants; (2) Multi-stakeholder provincial training by bringing together relevant members from DA, LGU, GAs, NGOs, Industry and academia to e.g., increase understanding of the need for new strains of seaweed from sexual reproduction, and progress to date; (3) establish community nursery that will act as model farms demonstrating the implementation of best practices because seeing is believing for small-scale fishers. Together with training this will help drive adoption. Seaweed-farmer scientists will be trained who can provide growth data on a monthly basis to help build up a clearer picture of seaweed growth rates for seedstock selection that will provide reliable and continuous source of high-quality seedlings for local community.

Rent expenses will cover costs that will be incurred during the sampling activities including accommodation, and boat and car rental during the sampling activities. Airfare costs are listed in the item for travel expenses.

Representation expenses will cover cost for food (lunch and snacks) in the annual meetings, and planning and evaluation meetings.

Subscription expenses will cover fees for access to scientific publications.

Services of a consultant in bookkeeping will be sought whenever necessary and services of a lawyer will also be sought when the need arises (e.g., when purchase of equipment is problematic due to unwanted practices of suppliers).

Other expenses include fees for inter-calibration purposes that may be required for some analytical methods and a budget for incidentals. Publication expenses will cover cost for page charges in journals with publication charges.

Indirect Cost Rate: UP-MSI

	BUDGET ITEM	Budget Requirement YR 1	YR 2
I. PERSONNEL SERVICES			
1	Personnel Salary	4,774,152.00	4,774,152.00
2	Honoraria	285,600.00	285,600.00
	Sub-Total for PS	**5,059,752.00**	**5,059,752.00**
II. MOOE			
1	Travel Expenses	3,400,000.00	2,653,000.00
2	Supplies and Materials	1,400,000.00	1,100,000.00
3	Communication Expenses	20,000.00	20,000.00
4	Repairs and Maintenance of Facilities	520,000.00	30,000.00
5	Transportation and Delivery Expenses	150,000.00	150,000.00
6	Utilities Expenses	-	-
7	Training Expenses	14,000,000.00	365,000.00
8	Membership Dues for the Institution	-	-
9	Advertising Expenses	-	-
10	Rent Expenses	50,000.00	200,000.00
11	Representation Expenses	30,000.00	30,000.00
12	Subscription Expenses	20,000.00	50,000.00
13	Survey Expenses	-	-
14	Professional Services	400,000.00	400,000.00
15	Taxes, Insurance Premiums and Other Fees	100,000.00	150,000.00
16	Other Maintenance and Operating Expenses	310,000.00	460,000.00
	Sub-Total for MOOE	**20,400,000.00**	**5,608,000.00**
III. Indirect Cost			
	Indirect for PS	1,011,950.40	1,011,950.40
	Indirect for MOOE	4,090,016.71	1,131,600.00
	Sub-Total Indirect Cost	**5,109,967.11**	**2,143,550.40**
IV. Equipment Outlay		**66,750,000.00**	
TOTAL PROPOSED BUDGET (UP-MSI)		**97,361,802.65**	**12,861,302.40**
TOTAL PROPOSED BUDGET FOR TWO YEARS (UP-MSI)			**110,223,105.05**

Equipment, MOOE and Justification for collaborating HEIs:

LINE ITEM BUDGET FOR MINDANAO STATE UNIVERSITY NAAWAN:
TOTAL PROPOSED BUDGET FOR 2 YEARS: 11,654,288.00

MSU Naawan will participate in research activities including sample collection, preservation for herbarium and chemical analyses, sample pre-treatment and analyses including anti-oxidative assays and basic biochemical profiling, which dictates the equipment being requested for this institution.

Personnel for PH Component: MSU Naawan

Position	Name	Annual Salary/Rate	Level of Effort	Cost for YR 1	Cost for YR 2
(1) HEI Collaborator	Rey Capangpangan	90,000.00	30%	90,000.00	90,000.00
(2-3) URA I	To be selected	795,120.00	100%	795,120.00	795,120.00
		TOTAL	**PhP**	**885,120.00**	**885,120.00**

JUSTIFICATION: Describe the role and responsibilities of each position.

(1): Dr. Rey Capangpangan of MSU Naawan will coordinate all research efforts at the partner HEIs including supervision of research staff based at the MSU Naawan.
(2) and (3) Two URAI: the two university research associates that will be identified from the pool of applicants (preferably a graduate student and hailing from nearby towns/provinces of the partner institutions) will assist in project activities from sampling to sample processing, data analysis, manuscript and report writing, and administrative tasks that will be necessary for smooth implementation of the project.

Equipment for MSU NAAWAN:

Item(s)	Cost/unit (PhP)	YR 1		YR 2	
		No. of Units	Sub-Total for YR 1	No. of Units	Sub-Total for YR 2
FT-IR and accessories	2,500,000.00	1	2,500,000.00		
HPLC with UV/RI detector and accessories	3,500,000.00	1	3,500,000.00		
UV-Vis spectrophotometer and accessories	700,000.00	1	700,000.00		
ultralow freezer and accessories	1,000,000.00	1	1,000,000.00		
furnace and accessories	450,000.00	1	450,000.00		
Rotary Evaporator with accessories	300,000.00		300,000.00		
TOTAL (PhP)			**8,450,000.00**		**0.00**

JUSTIFICATION: Describe the need or purpose of the equipment (e.g. how each unit will be used).

Equipment	Purpose	User	Location of the equipment
FT-IR and accessories	Will be used in the analyses of major organic groups present in an unknown compound to guide further purification steps.	For use of the research staff in MSU Naawan during the execution of the grant.	Will be housed and maintained at MSU NAAWAN.
HPLC with UV/RI detector and accessories	Will be used for qualitative and quantitative measurements of target analytes from samples.	For use of the research staff in MSU Naawan during the execution of the grant.	Will be housed and maintained at MSU NAAWAN.
UV-Vis spectrophotometer and accessories	Will be used for analyses of target compounds in samples.	For use of the research staff in MSU Naawan during the execution of the grant.	Will be housed and maintained at MSU NAAWAN.

Ultralow freezer and accessories	Will be used for sample storage especially samples that are thermally sensitive.	For use of the research staff in MSU Naawan during the execution of the grant.	Will be housed and maintained at MSU NAAWAN.
Furnace and accessories	Will be used for sample processing with focus on total carbon and nitrogen content.	For use of the research staff in MSU Naawan during the execution of the grant.	Will be housed and maintained at MSU NAAWAN.
Rotary Evaporator with accessories	Will be used for the removal of solvent under pressure during the extraction of target compounds from natural products.	For use of the research staff in MSU Naawan during the execution of the grant	Will be housed and maintained at MSU Naawan

Supplies and Materials for PH side: MSU NAAWAN

Item(s)	Cost/unit (PhP)	YR 1		YR 2	
		No. of units	Sub-Total for YR 1	No. of units	Sub-Total for YR 1
Extraction solvents	~2,000.00 to 5,000.00	~50	100,000.00	~50	100,000.00
HPLC grade solvents	~5,000.00 to 8,000.00	~25	150,000.00	~25	150,000.00
General laboratory chemicals	~500.00 to 2,500.00	~50	100,000.00	~50	100,000.00
Laboratory glassware/plastic ware and other materials	~500.00 to 5,000.00	~60	100,000.00	~60	100,000.00
TOTAL (PhP)			**450,000.00**		**450,000.00**

JUSTIFICATION:

Extraction solvents: Various extraction solvents (organic solvents or acidic/basic solvents will be used for extraction of desirable compounds from seaweeds. These solvents will be used in varying amounts and concentrations depending on the chemical properties of target compounds for extraction. Thus, the projected amount and cost are rough estimates only.

HPLC grade solvents: These solvents (acetonitrile, methanol, ultrapure water, acetone, and others) will be used in operating the HPLC for the precise separation and collection of compounds eluting from the HPLC. The costs and consumption of these solvents will be dependent on the nature of the target compounds for isolation.

General laboratory chemicals: these chemicals are required in sample pretreatment, clean-up, preconcentration, or cleaning steps in the laboratory.

Laboratory glassware/plastic ware and other materials: these are items needed as sample containers, reaction vessels, transfer and storage containers, and other routine laboratory procedures.

Indirect Cost Rate: MSU NAAWAN

	BUDGET ITEM	Budget Requirement YR 1	YR 2
I. PERSONNEL SERVICES			
1	Personnel Salary	795,120.00	795,120.00
2	Honoraria	90,000.00	90,000.00
	Sub-Total for PS	**885,120.00**	**885,120.00**
II. MOOE			
1	Travel Expenses		
2	Supplies and Materials	450,000.00	450,000.00
3	Communication Expenses		
4	Repairs and Maintenance of Facilities		
5	Transportation and Delivery Expenses		
6	Utilities Expenses		
7	Training Expenses		
8	Membership Dues for the Institution		
9	Advertising Expenses		

10	Rent Expenses		
11	Representation Expenses		
12	Subscription Expenses		
13	Survey Expenses		
14	Professional Services		
15	Taxes, Insurance Premiums and Other Fees		
16	Other Maintenance and Operating Expenses		
	Sub-Total for MOOE	**450,000.00**	**450,000.00**
	Indirect for PS	177,024.00	177,024.00
	Indirect for MOOE	90,000.00	90,000.00
	Sub-Total Indirect Cost	**267,024.00**	**267,024.00**
IV. Equipment Outlay		**8,450,000.00**	
TOTAL PROPOSED BUDGET: MSU NAAWAN		**10,052,144.00**	**1,602,144.00**
TOTAL PROPOSED BUDGET FOR 2 YEARS			**11,654,288.00**

LINE ITEM BUDGET FOR SORSOGON STATE UNIVERSITY:
TOTAL PROPOSED BUDGET FOR 2 YEARS: 12,674,288.00

Sorsogon State University will participate in research activities including sample collection, preservation for herbarium and chemical analyses, sample pre-treatment and analyses including taxonomic identification, molecular analyses, and morphological characterization, which dictates the equipment being requested for this institution.

Personnel for PH Component: Sorsogon State University

Position	Name	Annual Salary/Rate	Level of Effort	Cost for YR 1	Cost for YR 2
(1) HEI Collaborator	Richard Dumilag	90,000.00	30%	90,000.00	90,000.00
(2-3) URA I	To be selected	795,120.00	100%	795,120.00	795,120.00
		TOTAL	**PhP**	**885,120.00**	**885,120.00**

JUSTIFICATION: Describe the role and responsibilities of each position.

(1): Dr. Richard Dumilag of Sorsogon State University will coordinate all research efforts at the partner HEIs including supervision of research staff based at the SSU.
(2) and (3) Two URAI: the two university research associates that will be identified from the pool of applicants (preferably a graduate student and hailing from nearby towns/provinces of the partner institutions) will assist in project activities from sampling to sample processing, data analysis, manuscript and report writing, and administrative tasks that will be necessary for smooth implementation of the project.

Equipment for SSU:

Item(s)	Cost/unit (PhP)	YR 1		YR 2	
		No. of Units	Sub-Total for YR 1	No. of Units	Sub-Total for YR 2
Inverted microscope and accessories	1,000,000.00	1	1,000,000.00		
gradient PCR and accessories	600,000.00	1	600,000.00		
Compound microscope and accessories	300,000.00	1	300,000.00		
Stereomicroscope and accessories	300,000.00	1	300,000.00		
Gel Documentation System with complete parts and accessories	1,450,000.00	1	1,450,000.00		
bead ruptor tissue lyser and accessories	500,000.00	1	500,000.00		
liquid nitrogen containers and accessories	80,000.00	1	80,000.00		
refrigerator, freezer and accessories	100,000.00	2	200,000.00		
multiparameter and accessories for water sampling (data loggers)	1,250,000.00	1	1,250,000.00		
refrigerated centrifuge with accessories	1,200,000.00	1	1,200,000.00		
analytical balance and accessories	100,000.00	1	100,000.00		
toploading balance and accessories	50,000.00	1	50,000.00		
Digital Dry Bath with complete parts and accessories	90,000.00	1	90,000.00		
Pipettors with complete parts and accessories	95,000.00	4	380,000.00		
Vortex Mixer with complete parts and accessories	80,000.00	1	80,000.00		
Dead Air Cabinet with complete parts and accessories	450,000.00	1	450,000.00		

water purification system with complete parts and accessories	500,000.00	1	500,000.00		
Mini Centrifuge with complete parts and accessories	60,000.00	1	60,000.00		
DSLR camera and accessories	90,000.00	1	90,000.00		
Gel electrophoresis system with complete parts and accessories	70,000.00	1	70,000.00		
TOTAL (PhP)			8,750,000.00		0.00

JUSTIFICATION: Describe the need or purpose of the equipment (e.g. how each unit will be used).

Equipment	Purpose	User	Location of the equipment
Inverted microscope and accessories	Will be used for documentation especially looking at details of samples.	For use of the research staff in Sorsogon State University during the execution of the grant.	Will be housed and maintained at SSU.
gradient PCR and accessories	Will be used to determine presence of target genes.	For use of the research staff in Sorsogon State University during the execution of the grant.	Will be housed and maintained at SSU.
Compound microscope and accessories	Will be used for documentation especially looking at details of samples.	For use of the research staff in Sorsogon State University during the execution of the grant.	Will be housed and maintained at SSU.
Stereomicroscope and accessories	Will be used for documentation especially looking at details of samples.	For use of the research staff in Sorsogon State University during the execution of the grant.	Will be housed and maintained at SSU.
Gel Documentation System with complete parts and accessories	Will be used to check/visualize amplicons after gel electrophoresis	For use of the research staff in Sorsogon State University during the execution of the grant.	Will be housed and maintained at SSU.
bead ruptor tissue lyser and accessories	Required for sample preparation and pre-treatment.	For use of the research staff in Sorsogon State University during the execution of the grant.	Will be housed and maintained at SSU.
liquid nitrogen containers and accessories	Will be used for sample storage and preservation during transport.	For use of the research staff in Sorsogon State University during the execution of the grant.	Will be housed and maintained at SSU.
refrigerator, freezer and accessories	Will be used for sample storage and preservation.	For use of the research staff in Sorsogon State University during the execution of the grant.	Will be housed and maintained at SSU.
multiparameter for water sampling (data loggers)	Will be used for routine measurements of physico-chemical parameters during sampling.	For use of the research staff in Sorsogon State University during the execution of the grant.	Will be housed and maintained at SSU.
Refrigerated centrifuge with different rotors and accessories	Will be used for samples processing for pre-concentration or separation purposes.	For use of the research staff in Sorsogon State University during the execution of the grant.	Will be housed and maintained at SSU.
Analytical balance and accessories	Will be used for determination of sample weight.	For use of the research staff in Sorsogon State University during the execution of the grant.	Will be housed and maintained at SSU.
Toploading balance and accessories	Will be used for determination of sample weight.	For use of the research staff in Sorsogon State University during the execution of the grant.	Will be housed and maintained at SSU.
Digital Dry Bath with complete parts and accessories	Will be used for DNA extraction process	For use of the research staff in Sorsogon State University during the execution of the grant.	Will be housed and maintained at SSU.
Pipettors with complete parts and accessories	Will be used for various steps for sample preservation and other molecular downstream analysis (e.g. DNA extraction process, AGE, PCR, etc.)	For use of the research staff in Sorsogon State University during the execution of the grant.	Will be housed and maintained at SSU.
Vortex Mixer with complete parts and accessories	Will be used to mix and agitate samples during DNA extraction	For use of the research staff in Sorsogon State University during the execution of the grant.	Will be housed and maintained at SSU.

Dead Air Cabinet with complete parts and accessories	Will be used as a an axenic working station during molecular work	For use of the research staff in Sorsogon State University during the execution of the grant.	Will be housed and maintained at SSU.
Water purification system with complete parts and accessories	Will be used for producing ultrapure water	For use of the research staff in Sorsogon State University during the execution of the grant.	Will be housed and maintained at SSU.
Mini Centrifuge with complete parts and accessories	Will be used for a quick agitation and separation process during DNA extraction	For use of the research staff in Sorsogon State University during the execution of the grant.	Will be housed and maintained at SSU.
DSLR camera and accessories	For documentation	For use of the research staff in Sorsogon State University during the execution of the grant.	Will be housed and maintained at SSU.
Gel Electrophoresis System with complete parts and accessories	Will be used to separate DNA fragments (bands) and visualize amplicons	For use of the research staff in Sorsogon State University during the execution of the grant.	Will be housed and maintained at SSU.

E. Supplies and Materials:

Supplies and Materials for PH side: SSU

Item(s)	Cost/unit (PhP)	YR 1		YR 2	
		No. of units	Sub-Total for YR 1	No. of units	Sub-Total for YR 1
RNA extraction kits and other chemicals	~50,000.00	~6	300,000.00	~6	300,000.00
General laboratory chemicals	~5,000.00	~20	100,000.00	~20	100,000.00
Laboratory glassware/plastic ware and other materials	~2,000.00 to 5,000.00	~15	100,000.00	~15	100,000.00
TOTAL (PhP)			**500,000.00**		**500,000.00**

JUSTIFICATION:

RNA extraction kits and other chemicals: These kits are necessary for collection, preservation and analyses of RNA from samples.

General laboratory chemicals: these chemicals are required in sample pretreatment, clean-up, preconcentration, or cleaning steps in the laboratory.

Laboratory glassware/plastic ware and other materials: these are items needed as sample containers, reaction vessels, transfer and storage containers, and other routine laboratory procedures.

Indirect Cost Rate: SSU

	BUDGET ITEM	Budget Requirement YR 1	YR 2
I. PERSONNEL SERVICES			
1	Personnel Salary	795,120.00	795,120.00
2	Honoraria	90,000.00	90,000.00
	Sub-Total for PS	**885,120.00**	**885,120.00**
II. MOOE			
1	Travel Expenses		
2	Supplies and Materials	500,000.00	500,000.00
3	Communication Expenses		
4	Repairs and Maintenance of Facilities		
5	Transportation and Delivery Expenses		
6	Utilities Expenses		
7	Seaweed Farmers Training/Workshops Expenses	500,000.00	
8	Membership Dues for the Institution		
9	Advertising Expenses		

10	Rent Expenses		
11	Representation Expenses		
12	Subscription Expenses		
13	Survey Expenses		
14	Professional Services		
15	Taxes, Insurance Premiums and Other Fees		
16	Other Maintenance and Operating Expenses		
	Sub-Total for MOOE	**1,000,000.00**	**500,000.00**
	Indirect for PS	177,024.00	177,024.00
	Indirect for MOOE	200,000.00	100,000.00
	Sub-Total Indirect Cost	**377,024.00**	**277,024.00**
IV. Equipment Outlay		**8,750,000.00**	
TOTAL PROPOSED BUDGET: SSU		**11,012,144.00**	**1,662,144.00**
TOTAL PROPOSED BUDGET FOR 2 YEARS			**12,674,288.00**

Justification for MOOE #7 Seaweed Farmers Trainings/Workshop:
Farmer training (20-30 farmers per village in 5 villages per hub) focusing on practices that can minimise losses from disease and epiphytes and reduces marine plastic pollution. Costs will cover community organising/preparation for the workshops, seaweed farmer mobilisation, resource personnel to deliver training, training venue, materials, snacks/refreshments for participants

LINE ITEM BUDGET FOR UNIVERSITY OF EASTERN PHILIPPINES:
TOTAL PROPOSED BUDGET FOR 2 YEARS: 8,984,288.00

University of Eastern Philippines will participate in research activities including sample collection, preservation for herbarium and chemical analyses, sample pre-treatment and analyses including taxonomic identification, molecular analyses, and morphological characterization, which dictates the equipment being requested for this institution.

Personnel for PH Component: University of Eastern Philippines

Position	Name	Annual Salary/Rate	Level of Effort	Cost for YR 1	Cost for YR 2
(1) HEI Collaborator	Ronelie Salvador	90,000.00	30%	90,000.00	90,000.00
(2-3) URA I	To be selected	795,120.00	100%	795,120.00	795,120.00
		TOTAL	**PhP**	**885,120.00**	**885,120.00**

JUSTIFICATION: Describe the role and responsibilities of each position.

(1): Dr. Ronelie Salvador of University of Eastern Philippines will coordinate all research efforts at the partner HEIs including supervision of research staff based at the UEP.
(2) and (3) Two URAI: the two university research associates that will be identified from the pool of applicants (preferably a graduate student and hailing from nearby towns/provinces of the partner institutions) will assist in project activities from sampling to sample processing, data analysis, manuscript and report writing, and administrative tasks that will be necessary for smooth implementation of the project.

Equipment for UEP:

Item(s)	Cost/unit (PhP)	YR 1		YR 2	
		No. of Units	Sub-Total for YR 1	No. of Units	Sub-Total for YR 2
Inverted microscope and accessories	800,000.00	1	800,000.00		
stereomicroscope and accessories	300,000.00	1	300,000.00		
Oven and accessories	300,000.00	1	300,000.00		
Furnace and accessories	450,000.00	1	450,000.00		
analytical balance and accessories	100,000.00	1	100,000.00		
toploading balance and accessories	50,000.00	1	50,000.00		
UV-Vis spectrophotometer and accessories	700,000.00	1	700,000.00		
Water bath with temperature control and accessories	450,000.00	1	450,000.00		
Centrifuge with different rotors and accessories	1,200,000.00	1	1,200,000.00		
Multiparameter and accessories for water sampling	400,000.00	1	400,000.00		
pH meter, vortex mixer, hotplate, and accessories	150,000.00	1	150,000.00		
freezer and refrigerator for sample storage	100,000.00	1	100,000.00		
compound microscope and accessories	300,000.00	1	300,000.00		
Gopro and underwater camera and accessories	100,000.00	1	100,000.00		
data loggers and accessories	70,000.00	2	140,000.00		
TOTAL (PhP)			**5,540,000.00**		**0.00**

JUSTIFICATION: Describe the need or purpose of the equipment (e.g. how each unit will be used).

Equipment	Purpose	User	Location of the equipment
Inverted microscope and accessories	Will be used for documentation especially looking at details of samples.	For use of the research staff in University of Eastern Philippines during the execution of the grant.	Will be housed and maintained at UEP.
stereomicroscope and accessories	Will be used for documentation especially looking at details of samples.	For use of the research staff in University of Eastern Philippines during the execution of the grant.	Will be housed and maintained at UEP.
Oven and accessories	Will be used for sample drying and pretreatment procedures.	For use of the research staff in University of Eastern Philippines during the execution of the grant.	Will be housed and maintained at UEP.
Furnace and accessories	Will be used for sample processing with focus on total carbon and nitrogen content.	For use of the research staff in University of Eastern Philippines during the execution of the grant.	Will be housed and maintained at UEP.
analytical balance and accessories	Will be used for determination of sample weight.	For use of the research staff in University of Eastern Philippines during the execution of the grant.	Will be housed and maintained at UEP.
toploading balance and accessories	Will be used for determination of sample weight.	For use of the research staff in University of Eastern Philippines during the execution of the grant.	Will be housed and maintained at UEP.
UV-Vis spectrophotometer and accessories	Will be used for determination of target analytes using absorbance spectroscopy.	For use of the research staff in University of Eastern Philippines during the execution of the grant.	Will be housed and maintained at UEP.
Water bath with temperature control and accessories	Will be used for sample processing and pretreatment procedures.	For use of the research staff in University of Eastern Philippines during the execution of the grant.	Will be housed and maintained at UEP.
Centrifuge with different rotors and accessories	Will be used for samples processing for pre-concentration or separation purposes.	For use of the research staff in University of Eastern Philippines during the execution of the grant.	Will be housed and maintained at UEP.
Multiparameter and accessories for water sampling	Will be used for routine measurements of physico-chemical parameters during sampling.	For use of the research staff in University of Eastern Philippines during the execution of the grant.	Will be housed and maintained at UEP.
pH meter, vortex mixer, hotplate, and accessories	Wil be used for pH measurements, sample processing, and temperature-controlled experiments	For use of the research staff in University of Eastern Philippines during the execution of the grant.	Will be housed and maintained at UEP.
freezer and refrigerator and accessories for sample storage	Will be used for sample storage and preservation.	For use of the research staff in University of Eastern Philippines during the execution of the grant.	Will be housed and maintained at UEP.
compound microscope and accessories	Will be used for documentation especially looking at details of samples.	For use of the research staff in University of Eastern Philippines during the execution of the grant.	Will be housed and maintained at UEP.
Gopro and underwater camera and accessories	Will be used for documentation especially during sampling.	For use of the research staff in University of Eastern Philippines during the execution of the grant.	Will be housed and maintained at UEP.
data loggers and accessories	Will be used for data storage/collection.	For use of the research staff in University of Eastern Philippines during the execution of the grant.	Will be housed and maintained at UEP.

Supplies and Materials:

Supplies and Materials for PH side: UEP

Item(s)	Cost/unit (PhP)	YR 1		YR 2	
		No. of units	Sub-Total for YR 1	No. of units	Sub-Total for YR 1
Field and other materials	~1,000.00 to 5,000.00	~25	100,000.00	~25	100,000.00
General laboratory chemicals	~1,500.00 to 6,000.00	~20	100,000.00	~20	100,000.00
Laboratory glassware/plastic ware and other materials	~1,000.00 to 3,000.00	~40	100,000.00	~40	100,000.00
TOTAL (PhP)			**300,000.00**		**300,000.00**

JUSTIFICATION:

Field and other materials: The field and sampling materials, which include various items from ropes, non-slip mats, pails, etc., will be used for sample collection and processing while on the field.

General laboratory chemicals: these chemicals are required in sample pretreatment, clean-up, preconcentration, or cleaning steps in the laboratory.

Laboratory glassware/plastic ware and other materials: these are items needed as sample containers, reaction vessels, transfer and storage containers, and other routine laboratory procedures.

Indirect Cost Rate: UEP

	BUDGET ITEM	Budget Requirement	
		YR 1	YR 2
I. PERSONNEL SERVICES			
1	Personnel Salary	795,120.00	795,120.00
2	Honoraria	90,000.00	90,000.00
	Sub-Total for PS	**885,120.00**	**885,120.00**
II. MOOE			
1	Travel Expenses		
2	Supplies and Materials	300,000.00	300,000.00
3	Communication Expenses		
4	Repairs and Maintenance of Facilities		
5	Transportation and Delivery Expenses		
6	Utilities Expenses		
7	Seaweed Farmers Training/Workshops Expenses	500,000.00	
8	Membership Dues for the Institution		
9	Advertising Expenses		
10	Rent Expenses		
11	Representation Expenses		
12	Subscription Expenses		
13	Survey Expenses		
14	Professional Services		
15	Taxes, Insurance Premiums and Other Fees		
16	Other Maintenance and Operating Expenses		
	Sub-Total for MOOE	**800,000.00**	**300,000.00**

	Indirect for PS	177,024.00	177,024.00
	Indirect for MOOE	160,000.00	60,000.00
	Sub-Total Indirect Cost	**337,024.00**	**237,024.00**
IV. Equipment Outlay		**5,540,000.00**	
TOTAL PROPOSED BUDGET: UEP		**7,562,144.00**	**1,422,144.00**
TOTAL PROPOSED BUDGET FOR 2 YEARS			**8,984,288.00**

Justification for MOOE Item #7
Seaweed Farmers Training/ Workshops
Farmer training (20-30 farmers per village in 5 villages per hub) focusing on practices that can minimise losses from disease and epiphytes and reduces marine plastic pollution. Costs will cover community organising/preparation for the workshops, seaweed farmer mobilisation, resource personnel to deliver training, training venue, materials, snacks/refreshments for participants

LINE ITEM BUDGET FOR PARTIDO STATE UNIVERSITY:
TOTAL PROPOSED BUDGET FOR 2 YEARS: 12,054,288.00

Partido State University will participate in research activities including sample collection, preservation for herbarium and chemical analyses, sample pre-treatment and analyses including taxonomic identification, molecular analyses, and morphological characterization, which dictates the equipment being requested for this institution.

Personnel for PH Component: Partido State University

Position	Name	Annual Salary/Rate	Level of Effort	Cost for YR 1	Cost for YR 2
(1) HEI Collaborator	Michael Clores	90,000.00	30%	90,000.00	90,000.00
(2-3) URA I	To be selected	795,120.00	100%	795,120.00	795,120.00
		TOTAL	**PhP**	**885,120.00**	**885,120.00**

JUSTIFICATION: Describe the role and responsibilities of each position.

(1): Dr. Michael Clores of Partido State University will coordinate all research efforts at the partner HEIs including supervision of research staff based at the Partido SU.
(2) and (3) Two URAI: the two university research associates that will be identified from the pool of applicants (preferably a graduate student and hailing from nearby towns/provinces of the partner institutions) will assist in project activities from sampling to sample processing, data analysis, manuscript and report writing, and administrative tasks that will be necessary for smooth implementation of the project.

Equipment for PSU:

Item(s)	Cost/unit (PhP)	YR 1		YR 2	
		No. of Units	Sub-Total for YR 1	No. of Units	Sub-Total for YR 2
Oven drier and accessories	200,000.00	2	400,000.00		
analytical balance and accessories	100,000.00	1	100,000.00		
toploading balance and accessories	50,000.00	2	100,000.00		
furnace and accessories	450,000.00	1	450,000.00		
filtration system (vacuum pump, manifold, and accessories)	200,000.00	1	200,000.00		
UV-Vis spectrophotometer and accessories	700,000.00	1	700,000.00		
Spectrofluorometer and accessories	1,000,000.00	1	1,000,000.00		
Pump, motor with accessories	350,000.00	1	350,000.00		
data loggers and accessories	70,000.00	3	210,000.00		
scuba diving sets and accessories	80,000.00	2	160,000.00		
Compound microscope and accessories	300,000.00	2	600,000.00		
inverted microscope and accessories	800,000.00	1	800,000.00		
Stereomicroscope and accessories	300,000.00	1	300,000.00		
Mechanized rotating drum mixer and accessories	200,000.00	1	200,000.00		
Gopro and underwater camera and accessories	100,000.00	2	200,000.00		
tabletop centrifuge and accessories	150,000.00	1	150,000.00		
pH meter, vortex mixer, hotplate, and accessories	150,000.00	1	150,000.00		
Refractometer and accessories	100,000.00	1	100,000.00		
Dessicator and accessories	250,000.00	1	250,000.00		
electronic image analysis system and accessories	150,000.00	1	150,000.00		

shaker and accessories	500,000.00	1	500,000.00		
Multiparameter and accessories for water sampling	400,000.00	1	400,000.00		
refrigerated centrifuge with different rotors and accessories	1,500,000.00	1	1,500,000.00		
TOTAL (PhP)			**8,970,000.00**		**0.00**

JUSTIFICATION: Describe the need or purpose of the equipment (e.g. how each unit will be used).

Equipment	Purpose	User	Location of the equipment
Oven drier and accessories	Will be used for sample drying and pretreatment procedures.	For use of the research staff in Partido State University during the execution of the grant.	Will be housed and maintained at PSU.
analytical balance and accessories	Will be used for determination of sample weight.	For use of the research staff in Partido State University during the execution of the grant.	Will be housed and maintained at PSU.
toploading balance and accessories	Will be used for determination of sample weight.	For use of the research staff in Partido State University during the execution of the grant.	Will be housed and maintained at PSU.
furnace and accessories	Will be used for drying/carbon content dtermination.	For use of the research staff in Partido State University during the execution of the grant.	Will be housed and maintained at PSU.
filtration system (vacuum pump, manifold, and accessories)	Will be used for sample processing and pretreatment procedures.	For use of the research staff in Partido State University during the execution of the grant.	Will be housed and maintained at PSU.
UV-Vis spectrophotometer and accessories	Will be used for determination of target analytes using absorbance spectroscopy.	For use of the research staff in Partido State University during the execution of the grant.	Will be housed and maintained at PSU.
Spectrofluorometer and accessories	Will be used for determination of target analytes using fluorescence spectroscopy.	For use of the research staff in Partido State University during the execution of the grant.	Will be housed and maintained at PSU.
Pump, motor with accessories	Will be used for sampling.	For use of the research staff in Partido State University during the execution of the grant.	Will be housed and maintained at PSU.
data loggers and accessories	Will be used for data storage/collection during sampling.	For use of the research staff in Partido State University during the execution of the grant.	Will be housed and maintained at PSU.
scuba diving sets and accessories	Will be used for sampling.	For use of the research staff in Partido State University during the execution of the grant.	Will be housed and maintained at PSU.
Compound microscope and accessories	Will be used for documentation especially looking at details of samples.	For use of the research staff in Partido State University during the execution of the grant.	Will be housed and maintained at PSU.
inverted microscope and accessories	Will be used for documentation especially looking at details of samples.	For use of the research staff in Partido State University during the execution of the grant.	Will be housed and maintained at PSU.
Stereomicroscope and accessories	Will be used for documentation especially looking at details of samples.	For use of the research staff in Partido State University during the execution of the grant.	Will be housed and maintained at PSU.
Mechanized rotating drum mixer and accessories	Will be used for sample processing	For use of the research staff in Partido State University during the execution of the grant.	Will be housed and maintained at PSU.
Gopro and underwater camera and accessories	For documentation of samples and sampling processes	For use of the research staff in Partido State University during the execution of the grant.	Will be housed and maintained at PSU.
tabletop centrifuge and accessories	Will be used for sample processing and separation.	For use of the research staff in Partido State University during the execution of the grant.	Will be housed and maintained at PSU.

pH meter, vortex mixer, hotplate, and accessories	Wil be used for pH measurements, sample processing, and temperature-controlled experiments	For use of the research staff in Partido State University during the execution of the grant.	Will be housed and maintained at PSU.
Refractometer and accessories	For measurement of salinity during sampling	For use of the research staff in Partido State University during the execution of the grant.	Will be housed and maintained at PSU.
Dessicator and accessories	For use in drying and sample processing steps	For use of the research staff in Partido State University during the execution of the grant.	Will be housed and maintained at PSU.
electronic image analysis system and accessories	For analysis of images collated during sampling	For use of the research staff in Partido State University during the execution of the grant.	Will be housed and maintained at PSU.
shaker and accessories	Will be used for sample processing and pre-treatment steps	For use of the research staff in Partido State University during the execution of the grant.	Will be housed and maintained at PSU.
Multiparameter and accessories for water sampling	Will be used for routine measurements of physico-chemical parameters during sampling.	For use of the research staff in Partido State University during the execution of the grant.	Will be housed and maintained at PSU.
refrigerated centrifuge with different rotors and accessories	Will be used for samples processing for pre-concentration or separation purposes.	For use of the research staff in Partido State University during the execution of the grant.	Will be housed and maintained at PSU.

Supplies and Materials for PH side: PSU

Item(s)	Cost/unit (PhP)	YR 1		YR 2	
		No. of units	Sub-Total for YR 1	No. of units	Sub-Total for YR 1
Field and sample preservation materials	~1,000.00 to 5,000.00	~30	200,000.00	~30	200,000.00
General laboratory chemicals	~1,000.00 to 4,000.00	~50	100,000.00	~50	100,000.00
Laboratory glassware/plastic ware and other materials	~1,000.00 to 5,000.00	~40	100,000.00	~40	100,000.00
TOTAL (PhP)			**400,000.00**		**400,000.00**

JUSTIFICATION:

Field and sample preservation materials: The field and sampling materials, which include various items from ropes, non-slip mats, pails, etc., will be used for sample collection and processing while on the field.

General laboratory chemicals: these chemicals are required in sample pretreatment, clean-up, preconcentration, or cleaning steps in the laboratory.

Laboratory glassware/plastic ware and other materials: these are items needed as sample containers, reaction vessels, transfer and storage containers, and other routine laboratory procedures.

Indirect Cost Rate: PSU

	BUDGET ITEM	Budget Requirement YR 1	YR 2
I. PERSONNEL SERVICES			
1	Personnel Salary	795,120.00	795,120.00
2	Honoraria	90,000.00	90,000.00
	Sub-Total for PS	**885,120.00**	**885,120.00**
II. MOOE			
1	Travel Expenses		
2	Supplies and Materials	400,000.00	400,000.00

Budget and Justification: PhycoPRO – UCSD

A. Personnel: List employee(s) (including names for each identified position) of the applicant/recipient organization, including in-kind costs for those positions whose work is tied to the grant project.

Position	Name	Annual Salary/Rate	Level of Effort	Cost for Year 1	Cost for Year 2	Total Cost for 2 Years
(1) Collaborator	William Gerwick	105,600.00	12.5%	2,922,536.82	3,030,961.94	5,953,525.76
(2) Collaborator	Lena Gerwick	90,000.00	12.5%	1,026,447.84	1,060,097.90	2,086,545.74
(3) Postdoctoral Fellow	Harpreet Singh	90,000.00	100%	3,667,069.56	3,986,043.60	7,653,113.16
(4) SRA	Evgenia Glukhov	549,360.00	16.67%	877,198.05	917,784.86	1,794,979.92
			TOTAL	8,493,276.27	8,994,888.31	17,488,164.58

JUSTIFICATION: Describe the role and responsibilities of each position.

(1) and (2): Both collaborators, William and Lena Gerwick, will provide general research directions and supervision of research staff based at the Scripps Institution of Oceanography. The expertise of William and Lena in natural products chemistry and bio-assay guided isolation and elucidation of chemical constituents of seaweeds and associated epiphytes will be critical for the success of the project. It is also expected that they will impart valuable strategies in terms of experimental design pertaining to isolation and chemical elucidation of bioactive compounds. Both William and Lena are considered key staff.

(3) : The Postdoctoral fellow will perform most of the research activities pertaining to isolation and chemical elucidation of potential bioactive compounds from seaweed extracts and will be considered key staff.

(4) : The SRA will assist in all research activities and will be considered key staff. In addition to research activities, the SRA will also assist in administrative tasks and coordination of project deliverables with the PH-based project team.

Key staff positions require prior approval after review of credentials of resume and job description.

B. Travel : Explain need for all travel other than that required by this application. Relevant Philippine government policies for travel such as Executive Order 77, series of 2019 on Official Local and Foreign Travels shall apply.

NONE

C. Equipment (for Philippines side only):

NONE

D. Supplies and Materials: UCSD

Item(s)	Cost/unit (PhP)	YR 1		YR 2	
		No. of units	Sub-Total for YR 1	No. of units	Sub-Total for YR 1
Reagents and materials for sample identification and preliminary testing	~13,750.00	~75	825,000.00		
Chemicals for analysis and evaluation of extracts for activity and other parameters	~11,183.33	~75	671,000.00		
Certified reference materials for identification of isolated compounds with activity	~38,500.00	~15	577,500.00		
Reagents for sample analyses of various parameters	~6,600.00			~150	990,000.00
Chemicals and reagents for analysis of seaweed extracts	~5,516.14			~150	827,420.00
TOTAL (PhP)			**2,073,500.00**		**1,817,420.00**

JUSTIFICATION:

Reagents and materials for sample identification and preliminary testing will include various solvents, test kits, and use of sophisticated equipment for sample processing. The costs are projected for assays/analyses of about 75 samples for screening.

Chemicals for analysis and evaluation of extracts for activity and other parameters will include all chemicals (standards, assay kits, chemicals for quality control checks) and consumables for use in instruments for sample processing. The costs are estimated for about 75 purified or pre-concentrated extracts for bioassay guided isolation and further activity screening.

Certified reference materials will include all costs for procurement of compounds with known activity or identified compounds with similar activity for comparison of retention times or mass spectra and will serve as surrogate or internal standards during chromatographic separations and chemical structure elucidation and bioactivity screening (for positive or negative controls).

Reagents for sample analyses of various parameters will include all general laboratory chemicals for routine isolation, separation, and purification or preconcentration of potential bioactive compounds from seaweeds and associated epiphytes following an optimized protocol. The costs are estimated for about 150 samples that will be processed for the screening.

Chemicals and reagents for analysis of seaweed extracts will include all solvents, assay kits, and other consumables needed for routine screening of purified or preconcentrated extracts from seaweed extracts. The costs are estimated for about 150 extracts that will be processed for the screening.

E. Professional Services: These include costs of services such as legal, consultancy, environment/sanitary, general, and other professional services required by the project. A contractual arrangement may be entered into to engage the services of service providers to carry out a portion of the project effort or for the acquisition of routine goods or services under the project. Such arrangements may be in the form of consortium agreements or contracts. A consultant is an individual retained to provide professional advice or services for a fee. All procurement transactions shall be conducted in a manner to provide to the maximum extent practical, open and free competition.

NONE

F. Construction: NOT ALLOWED

G. Maintenance and Other Operating Expenses (MOOE) may include the following expenses not covered in any of the previous budget categories

Item(s)	Cost for YR 1	Cost for YR 2	TOTAL
Communication expenses			
Communication expenses during project implementation	147,015.00	157,850.00	204,865.00
Repairs and Maintenance of Facilities			
Maintenance of clean room			
Renovation cost			
Transportation and Delivery Expenses			
Freight costs			
Training Expenses			
Membership Dues for the Institution			
		-	
Rent Expenses			
Boats for sampling			
Representation Expenses			
Subscription expenses			
Professional fees			
Taxes, Insurance Premiums, and Other Fees			
Other Maintenance and Operating Expenses			
	487,740.00	715,000.00	1,202,740.00
TOTAL	**634,755.00**	**872,850.00**	**1,507,605.00**

JUSTIFICATION:

Communication expenses will cover cost for availing of internet connectivity and purchasing of SIM and prepaid cards for use of project staff in coordinating sampling activities especially during travel to the Philippines of the UCSD research staff.

Other expenses include fees for access to mass spectral libraries, library/databases of bioactivity and biochemical properties, and inter-calibration purposes that may be required for some analytical methods and a budget for incidentals.

H. Administrative Overhead Cost (AOC):
A maximum fixed AOC of 20% of total PS and MOOE items shall apply.

Budget Justification
Project Title: Lakas 2021-035: PHYTOCHEMICAL CHARACTERIZATION OF MACROALGAE FOR FOOD AND HIGH VALUE PRODUCTS (PHYCOPRO)

BUDGET SUMMARY:

BUDGET ITEM	BUDGET REQUIREMENT		
	YR 1 ___	YR 2	TOTAL
I. PS	8,493,276.27	8,994,888.31	17,488,164.58
II. MOOE	2,708,225.00	2,690,270.00	1,507,605.00
III. EO	0.00	0.00	0.00
IV. AOC	2,240,306.25	2,337,031.66	4,577,337.93
TOTAL PROPOSED BUDGET	13,441,837.53	14,022,189.97	27,464,027.49

Line Item Budget

COMMISSION ON HIGHER EDUCATION

LINE ITEM BUDGET: UP-MSI

Project Title: **Phytochemical characterization of macroalgae for food and high value products (PhycoPRO)**
Project Duration: TWO Years
Implementing Institution/s: UP Marine Science Institute,Mindanao State University-Naawan,Partido State University, Sorsogon State University, and University of Eastern Philippines

	ITEMS		Total
I.	**PERSONNEL SERVICES (PS)**		
	Direct Cost	P	10,119,504.00
	Sub-total for PS	P	
II.	**MAINTENANCE AND OTHER OPERATING EXPENSES (MOOE)**		
	Direct Cost		
	Travel Expenses		6,053,000.00
	Supplies and Materials		2,500,000.00
	Communication Expenses		40,000.00
	Repairs and Maintenance of Facilities		550,000.00
	Transportation and Delivery Expenses		300,000.00
	Utilities Expenses		
	Training Expenses		14,365,000.00
	Membership Dues for the Institution		
	Rent Expenses		250,000.00
	Representation Expenses		160,083.54
	Subscription Expenses		70,000.00
	Survey Expenses		
	Professional Services		800,000.00
	Taxes, Insurance Premiums and Other Fees		250,000.00
	Other Maintenance and Operating Expenses		770,000.00
	Instrument fee for inter-calibration purposes (120,000)		
	Publication expenses (650,000)		
	Sub-total for MOOE	P	**26,108,083.54**
III.	**EQUIPMENT OUTLAY (EO)** - *all in: include customs duties, etc*		
		P	66,750,000.00
	Sub-total for EO	P	**66,750,000.00**
IV.	**ADMINISTRATIVE OVERHEAD COST (AOC)**		
	AOC for PS	P	2,023,900.80
	AOC for MOOE		5,221,616.71
	Sub-total for AOC	P	**7,245,517.51**
	GRAND TOTAL	P	**110,223,105.05**

Prepared and Submitted by:

Michael Y. Roleda, Dr. rer. nat.
(Project Leader)

Irene B. Rodriguez, Dr. rer. nat.
(Co-Project Leader)

Line Item Budget

COMMISSION ON HIGHER EDUCATION

LINE ITEM BUDGET: MSUN

Project Title: **Phytochemical characterization of macroalgae for food and high value products (PhycoPRO)**
Project Duration: **TWO Years**
Implementing Institution/s: UP Marine Science Institute,Mindanao State University-Naawan,Partido State University, Sorsogon State University, and University of Eastern Philippines

	ITEMS		Total
I.	**PERSONNEL SERVICES (PS)**		
	Direct Cost	P	1,770,240.00
	Sub-total for PS	P	
II.	**MAINTENANCE AND OTHER OPERATING EXPENSES (MOOE)**		
	Direct Cost		
	Travel Expenses		
	Supplies and Materials		900,000.00
	Communication Expenses		
	Repairs and Maintenance of Facilities		
	Transportation and Delivery Expenses		
	Utilities Expenses		
	Training Expenses		
	Membership Dues for the Institution		
	Rent Expenses		
	Representation Expenses		
	Subscription Expenses		
	Survey Expenses		
	Professional Services		
	Taxes, Insurance Premiums and Other Fees		
	Other Maintenance and Operating Expenses		
	(Provide account)		
	Sub-total for MOOE	P	**900,000.00**
III.	**EQUIPMENT OUTLAY (EO)** - *all in; include customs duties, etc*		
		P	8,450,000.00
	Sub-total for EO	P	**8,450,000.00**
IV.	**ADMINISTRATIVE OVERHEAD COST (AOC)**		
	AOC for PS	P	354,048.00
	AOC for MOOE		180,000.00
	Sub-total for AOC	P	**534,048.00**
	GRAND TOTAL	P	**11,654,288.00**

Prepared and Submitted by:

Rey Y. Capangpangan, Ph.D.
(Project Collaborator, MSU-Naawan)

Line Item Budget

COMMISSION ON HIGHER EDUCATION

LINE ITEM BUDGET: ParSU

Project Title: Phytochemical characterization of macroalgae for food and high value products (PhycoPRO)
Project Duration: TWO Years
Implementing Institution/s: UP Marine Science Institute,Mindanao State University-Naawan,Partido State University, Sorsogon State University, and University of Eastern Philippines

	ITEMS		Total
I.	**PERSONNEL SERVICES (PS)**		
	Direct Cost	P	1,770,240.00
	Sub-total for PS	P	
II.	**MAINTENANCE AND OTHER OPERATING EXPENSES (MOOE)**		
	Direct Cost		
	Travel Expenses		
	Supplies and Materials		800,000.00
	Communication Expenses		
	Repairs and Maintenance of Facilities		
	Transportation and Delivery Expenses		
	Utilities Expenses		
	Training Expenses		
	Membership Dues for the Institution		
	Rent Expenses		
	Representation Expenses		
	Subscription Expenses		
	Survey Expenses		
	Professional Services		
	Taxes, Insurance Premiums and Other Fees		
	Other Maintenance and Operating Expenses		
	(Provide account)		
	Sub-total for MOOE	P	**800,000.00**
III.	**EQUIPMENT OUTLAY (EO)** - all in; include customs duties, etc		
		P	8,450,000.00
	Sub-total for EO	P	**8,970,000.00**
IV.	**ADMINISTRATIVE OVERHEAD COST (AOC)**		
	AOC for PS	P	354,048.00
	AOC for MOOE		160,000.00
	Sub-total for AOC	P	**514,048.00**
	GRAND TOTAL	P	**12,054,288.00**

Prepared and Submitted by:

Michael A. Clores, Ph.D.
(Project Collaborator, ParSU)

Line Item Budget

COMMISSION ON HIGHER EDUCATION

LINE ITEM BUDGET: SorSU

Project Title: Phytochemical characterization of macroalgae for food and high value products (PhycoPRO)
Project Duration: TWO Years
Implementing Institution/s: UP Marine Science Institute,Mindanao State University-Naawan,Partido State University, Sorsogon State University, and University of Eastern Philippines

	ITEMS		Total
I.	**PERSONNEL SERVICES (PS)**		
	Direct Cost	P	1,770,240.00
	Sub-total for PS	P	1,770,240.00
II.	**MAINTENANCE AND OTHER OPERATING EXPENSES (MOOE)**		
	Direct Cost		
	Travel Expenses		
	Supplies and Materials		1,000,000.00
	Communication Expenses		
	Repairs and Maintenance of Facilities		
	Transportation and Delivery Expenses		
	Utilities Expenses		
	Training Expenses		500,000.00
	Membership Dues for the Institution		
	Rent Expenses		
	Representation Expenses		
	Subscription Expenses		
	Survey Expenses		
	Professional Services		
	Taxes, Insurance Premiums and Other Fees		
	Other Maintenance and Operating Expenses		
	(Provide account)		
	Sub-total for MOOE	P	**1,500,000.00**
III.	**EQUIPMENT OUTLAY (EO)** - all in; include customs duties, etc		
		P	8,450,000.00
	Sub-total for EO	P	**8,750,000.00**
IV.	**ADMINISTRATIVE OVERHEAD COST (AOC)**		
	AOC for PS	P	354,048.00
	AOC for MOOE		300,000.00
	Sub-total for AOC	P	**654,048.00**
	GRAND TOTAL	P	**12,674,288.00**

Prepared and Submitted by:

Richard V. Dumilag, Ph.D.
(Project Collaborator, SorSU)

Line Item Budget

COMMISSION ON HIGHER EDUCATION

LINE ITEM BUDGET: UEP

Project Title: Phytochemical characterization of macroalgae for food and high value products (PhycoPRO)
Project Duration: TWO Years
Implementing Institution/s: UP Marine Science Institute

	ITEMS		Total
I.	**PERSONNEL SERVICES (PS)**		
	Direct Cost	P	1,770,240.00
	Sub-total for PS	P	
II.	**MAINTENANCE AND OTHER OPERATING EXPENSES (MOOE)**		
	Direct Cost		
	Travel Expenses		
	Supplies and Materials		600,000.00
	Communication Expenses		
	Repairs and Maintenance of Facilities		
	Transportation and Delivery Expenses		
	Utilities Expenses		
	Training Expenses		500,000.00
	Membership Dues for the Institution		
	Rent Expenses		
	Representation Expenses		
	Subscription Expenses		
	Survey Expenses		
	Professional Services		
	Taxes, Insurance Premiums and Other Fees		
	Other Maintenance and Operating Expenses		
	(Provide account)		
	Sub-total for MOOE	P	1,100,000.00
III.	**EQUIPMENT OUTLAY (EO)** - *all in: include customs duties, etc*		
		P	8,450,000.00
	Sub-total for EO	P	**5,540,000.00**
IV.	**ADMINISTRATIVE OVERHEAD COST (AOC)**		
	AOC for PS	P	354,048.00
	AOC for MOOE		220,000.00
	Sub-total for AOC	P	**574,048.00**
	GRAND TOTAL	P	8,984,288.00

Prepared and Submitted by:

Ronelie C. Salvador, Ph.D.
(Project Collaborator, UEP)

Narrative Proposal
Project Results Workplan/Matrix

Project Title: Phytochemical characterization of macroalgae for food and high value products (PhycoPRO)

Total Funded Duration: TWO Years

CALENDAR YEAR AND QUARTER			YEAR 1				YEAR 2			
Project Duration (in months per quarter; for 24 months)			Q1	Q2	Q3	Q4	Q1	Q2	Q3	Q4
OBJECTIVES	**EXPECTED OUTPUT**	**Metrics used to track and measure progress of work leading to output.**	**Quantified relevant % progress**							
Part 1 – Planning										
1.1 Formally initiate project with kickoff meeting	Detailed project work plan	Project plan with details of deliverables.	100%							
1.2 Build project team	Team staffing report	Hiring of research team members	100%							
1.3 Procure equipment	Equipment purchased	Percentage of equipment purchased	10%	50%	75%	100%				
1.4 Procure chemicals, laboratory and office supplies	Chemicals, laboratory and office supplies purchased	Percentage of chemicals, laboratory and office supplies purchased	10%	50%	75%	100%				
1.5 Identify sampling sites and establish local contacts	Sampling sites and local contacts identified	List of sampling sites and local contacts	100%							
1.6 Obtain necessary permits/licenses from pertinent government agencies	Permits/licenses acquired	List of licenses/permits obtained	50%	100%						
1.7 Document progress accomplishment	Annual progress report	Delivery of annual progress report				100%				100%
Part 2 – Seaweed collection for herbarium collection and method optimization										
2.1 Seaweed collection for optimization of methods	Seaweed samples collected	Report on seaweed collection	25%	100%						
2.2 Deposit samples in herbarium	Seaweed samples deposited in herbarium	Report on materials deposited in herbarium		100%						
2.3 Pre-treatment of samples for method optimization	Pre-treatment methods optimized and validated	Report on pre-treatment methods								

Part 3 – Optimization and validation of methods										
3.1 Optimize method for proximal analysis	Optimized method for proximal analysis	Report on validated method		50%	100%					
3.2 Optimize method for total lipid, fucoxanthin, and fucosterol analyses	Optimized method for lipid, fucoxanthin, and fucosterol analyses	Report on validated method		30%	60%	80%	100%			
3.3 Optimize method for fatty acid extraction and analysis	Optimized method for fatty acid extraction and analysis	Report on validated method		30%	60%	80%	100%			
3.4 Optimize method for amino acid analysis and specific component MAAs	Optimized method for amino acid and targeted MAAs analyses	Report on validated method		30%	60%	80%	100%			
3.5 Optimize method for β-carotene extraction and analysis	Optimized method for β-carotene extraction and analysis	Report on validated method		30%	60%	80%	100%			
3.6 Optimize method for iodine analysis and estimation of iodine intake from seaweeds	Optimized method for determination of iodine content and intake from seaweeds	Report on validated method			30%	60%	80%	100%		
3.7 Optimize method for analysis of phenolic content	Optimized method for analysis of phenolic content	Report on validated method			30%	60%	80%	100%		
3.8 Optimize method for determination of antioxidant activity	Optimized method for antioxidant activity determination	Report on validated method			30%	60%	80%	100%		
3.9 Optimize method for total metal analysis	Optimized method for trace metal determination	Report on validated method			30%	60%	80%	100%		
3.10 Optimize method for speciation analysis of specific trace metals of interest	Optimized method for speciation analysis of target trace metals	Report on validated method			30%	60%	80%	100%		
3.11 Optimize in-vitro dialyzability procedure to determine bioavailability and bioavailability of trace metals	Optimized method for in-vitro dialyzability procedure	Report on validated method			30%	60%	80%	100%		
Part 4 – Screening of seaweed and associated epiphytes for bioactivities										
4.1 Training of staff for seaweed handling and preservation for drug discovery related studies	Research staff trained	Report on training conducted	25%	100%						

4.2 Extraction for drug discovery related work, sending of materials to CA	Materials sent to CA collaborator	Report on materials sent to CA collaborator		25%	50%	100%				
4.3 Analysis and evaluation of extracts for activity and other parameters	Extracts evaluated for bioactivity	Results incorporated in manuscript draft				50%			100%	
4.4 Identification of isolated compounds with activity	Compounds identified or characterized	Results incorporated in manuscript draft				25%		50%		100%
4.5 Training of staff for isolation, purification, and identification of potential compounds	Staff trained in relevant processes	Protocols for relevant processes prepared				25%				100%
Part 5 – Seaweed collection and evaluation of phytochemical parameters										
5.1 Sampling for identification and preliminary testing	Seaweed samples collected	Report on Seaweed species identification		50%	100%					
5.2 Deposit samples in herbarium	Seaweed samples deposited in herbarium	Report on materials deposited in herbarium		100%						
5.3 Sampling for analysis of various parameters	Seaweed samples collected	Report on seaweed species collected					20%	50%	100%	
5.4 Analyses of seaweed and seaweed extracts using optimized methods	Various parameters analyzed in seaweed samples	Report on data acquired						20%	80%	100%
5.5 Data processing and analyses	Data collected	Report on data acquired						10%	70%	100%
Part 6 – Training/Workshop on seaweed farming										
6.1 Preparation of seaweed farming information materials	Information materials prepared	Prepared information materials for dissemination	25%	75%	100%					
6.2 Conduct of training/workshop for seaweed farmers	Training/workshops carried out with partner HEIs for seaweed farmers	Report on the training/workshop conducted		10%	20%	40%	60%	80%	100%	
Part 7 – Post-implementation										
7.1 Write research papers	Research papers	Number of research papers							50%	100%
7.2 Project evaluation	Final project report	Final project report								100%